

Wire ID	Description	Type	System
POWER (100s)			
101-P-PWR-LPO-BMS-XT9	LiPo packs to BMS	P	PWR
102-P-PWR-BMS-FUS-IN	BMS output to fuse	P	PWR
103-P-PWR-PWB-FUS-BA+		P	PWR
104-P-PWR-PWB-XT9-BA-	Fuse to kill switch SKS	P	PWR
105-P-PWR-PWB-MKS-MKS	MKS kill switch to power board	P	PWR
106-P-PWR-PWB-SKS-SKS	SKS kill switch to power board	P	PWR
107-P-PWR-PWB-BKI-CIN	Power board 24V to Buck 24→12V	P	PWR
108-P-PWR-PWB-BKO-COT	Power board 24V to Buck 24→5V	P	PWR
109-P-PWR-PWB-FAN-J15	Power board to fans (J15)	P	PWR
110-P-PWR-PWB-FAN-J16	Power board to fans (J16)	P	PWR
DRIVE (200s)			
201-P-DRV-PWB-SPL-RKL	Power board J17 Rocker L to splitter	P	DRV
202-P-DRV-PWB-SPL-RKR	Power board J18 Rocker R to splitter	P	DRV
203-P-DRV-SPL-BRL-CAN	Splitter to left brushless controller	P	DRV
204-P-DRV-SPL-BRR-CAN	Splitter to right brushless controller	P	DRV
205-P-DRV-SPL-BRF-CAN	Splitter to front brushless controller	P	DRV
206-P-DRV-BRL-MOL-IN	Left brushless to left drive motor	P	DRV
207-P-DRV-BRR-MOR-IN	Right brushless to right drive motor	P	DRV
208-P-DRV-BRF-MOF-IN	Front brushless to front drive motor	P	DRV
209-P-DRV-PWB-HD1-HDL	Power board J26 to headlight 1	P	DRV
210-P-DRV-HD1-HD2-IN	Headlight 1 daisy-chain to headlight 2	P	DRV
211-P-DRV-HD2-HD3-IN	Headlight 2 daisy-chain to headlight 3	P	DRV
212-D-DRV-PWB-SWT-RKS	Power board J23 RKS to switches	D	DRV
COMMUNICATIONS (300s)			
301-D-COM-PWB-CAN-J27	Power board CAN J27 to CANable	D	COM
302-P-COM-PWB-NSW-NSW	Buck 24→12V to network switch	P	COM
303-D-COM-NSW-JET-ETH	Network switch to Jetson ethernet	D	COM
304-D-COM-NSW-POE-ETH	Network switch to POE injector	D	COM
305-D-COM-POE-ANT-ETH	POE injector to antenna	D	COM
306-D-COM-JET-HUB-USB	Jetson to USB hub	D	COM
307-D-COM-HUB-CAN-USB	USB hub to CANable	D	COM
308-D-COM-SPL-JET-CAN	Drive splitter CAN to Jetson	D	COM
309-P-COM-PWB-JET-JET	Power board to Jetson	D	COM
310-P-COM-PWB-POE-ANT	Power board to ANTENNA	D	COM
311-P-COM-PWB-URT-RKS	Power board UART to UART board	D	COM
312-D-COM-ABX-RPI-CAN	Power board J7 RBPI ARM to Raspberry Pi	D	COM
313-D-COM-RPI-HAT-GPIO	Raspberry Pi to Pi Hat	D	COM
314-P-COM-PWB-PAN-PAN	Power board GPS+PanTilt J16 to pantilt	D	COM
315-P-COM-PAN-PIT-SRV	Pantilt to pitch servo	D	COM
316-P-COM-PAN-YAW-SRV	Pantilt to yaw servo	D	COM
317-P-COM-PAN-TOF-TOF	Pantilt to ToF sensor	D	COM
ARM (400s)			
401-P-ARM-PWB-APD-J7	Power board J7 to arm PD board	P	ARM
402-P-ARM-APD-ABX-24V	Arm PD board 24V to arm box board	P	ARM

403-P-ARM-APD-RPI-PWR	Arm PD board to Raspberry Pi	P	ARM
404-D-ARM-ABX-ELB-ELB	Arm box board to elbow ESC	D	ARM
405-D-ARM-ABX-SHO-SHO	Arm box board to shoulder ESC	D	ARM
406-D-ARM-ABX-WST-WST	Arm box board to waist ESC	D	ARM
401-P-ARM-ABX-ABR-PWR	Arm box board BRUSHED_PW to arm brushed board	D	ARM
402-P-ARM-ABR-GRM-GRP	Arm brushed board to gripper motor	P	ARM
403-P-ARM-ABR-WPM-WRP	Arm brushed board to wrist pitch motor	P	ARM
404-P-ARM-ABR-WRM-WRR	Arm brushed board to wrist roll motor	P	ARM
405-D-ARM-ABR-GRE-GRE	Arm brushed board to gripper encoder	D	ARM
406-D-ARM-ABR-WPE-WPE	Arm brushed board to wrist pitch encoder	D	ARM
407-D-ARM-ABR-WRE-WRE	Arm brushed board to wrist roll encoder	D	ARM
408-D-ARM-ABR-LS1-LIM	Arm brushed board to limit switch 1-2	D	ARM
409-D-ARM-ABR-LS3-LIM	Arm brushed board to limit switch 3-4	D	ARM
410-D-ARM-ABR-LS5-LIM	Arm brushed board to limit switch 5-6	D	ARM
411-D-ARM-ABR-TOF	Arm brushed board to ToF sensor (arm)	D	ARM
412-D-ARM-ABR-LAS	Arm brushed board to laser	D	ARM
413-P-ARM-ABX-APD-24V	Arm Box Board RPI 24V to Arm PD Board	P	ARM
414-P-ARM-APD-RPI-RPI		P	ARM
SCIENCE (500s)			
501-P-SCI-PWB-SCA-SCA	Power board Arm/Sci J22 to science board	D	SCI

End 1 — Device	End 1 — Connector	End 2 — Device	End 2 — Connector	Length
LiPo Pack	XT90	BMS	XT90	TBD
BMS	OUT	Fuse	IN	TBD
Fuse	OUT	Kill Switch (MKS)	IN	TBD
Fuse	OUT	Kill Switch (SKS)	IN	TBD
Kill Switch (MKS)	OUT	Power Board	BAT+	TBD
Kill Switch (SKS)	OUT	Power Board	BAT-	TBD
Power Board	24V	Buck Conv 24→12V	24VIN	TBD
Power Board	24V	Buck Conv 24→5V	24VIN	TBD
Power Board	J15	Fans	PWR	TBD
Power Board	J16	Fans	PWR	TBD
Power Board	RKL	Splitter	IN	TBD
Power Board	RKR	Splitter	IN	TBD
Splitter	OUT	Brushless Ctrl (L)	CAN	TBD
Splitter	OUT	Brushless Ctrl (R)	CAN	TBD
Splitter	OUT	Brushless Ctrl (F)	CAN	TBD
Brushless Ctrl (L)	OUT	Drive Motor (L)	IN	TBD
Brushless Ctrl (R)	OUT	Drive Motor (R)	IN	TBD
Brushless Ctrl (F)	OUT	Drive Motor (F)	IN	TBD
Power Board	HDL	Headlight 1	IN	TBD
Headlight 1	OUT	Headlight 2	IN	TBD
Headlight 2	OUT	Headlight 3	IN	TBD
Power Board	RKS	Switches	IN	TBD
Power Board	J27	CANable	USB	TBD
Buck Conv 24→12V	12VOUT	Network Switch	PWR	TBD
Network Switch	ETH	Jetson	ETH	TBD
Network Switch	ETH	POE	IN	TBD
POE	OUT	Antenna	ETH	TBD
Jetson	USB	USB Hub	USB	TBD
USB Hub	USBA	CANable	USBC	TBD
Splitter	CAN	Jetson	CAN	TBD
Power Board	GPS1	GPS 1	IN	TBD
Power Board	GPS2	GPS 2	IN	TBD
Power Board	UART	UART Board (Sonia)	IN	TBD
Power Board	J7	Raspberry Pi	GPIO	TBD
Raspberry Pi	GPIO	Raspberry Pi Hat	GPIO	TBD
Power Board	J16	Pantilt	PWR	TBD
Pantilt	SRV	Pitch Servo	IN	TBD
Pantilt	SRV	Yaw Servo	IN	TBD
Pantilt	SIG	ToF Sensor (Pan)	IN	TBD
Power Board	J7	Arm PD Board	PWR	TBD
Arm PD Board	24V	Arm Box Board	PWR	TBD

Arm PD Board	PWR	Raspberry Pi	PWR	TBD
Arm Box Board	ELB	Elbow ESC	IN	TBD
Arm Box Board	SHO	Shoulder ESC	IN	TBD
Arm Box Board	WST	Waist ESC	IN	TBD
Arm Box Board	CAN	Arm Brushed Board	CAN	TBD
Arm Brushed Board	GRP	Gripper Motor	IN	TBD
Arm Brushed Board	WPT	Wrist Pitch Motor	IN	TBD
Arm Brushed Board	WRL	Wrist Roll Motor	IN	TBD
Arm Brushed Board	ENC	Gripper Encoder	IN	TBD
Arm Brushed Board	ENC	Wrist Pitch Encoder	IN	TBD
Arm Brushed Board	ENC	Wrist Roll Encoder	IN	TBD
Arm Brushed Board	LIM	Limit Switch 1	IN	TBD
Arm Brushed Board	LIM	Limit Switch 2	IN	TBD
Arm Brushed Board	LIM	Limit Switch (Grip)	IN	TBD
Arm Brushed Board	SIG	ToF Sensor (Arm)	IN	TBD
Arm Brushed Board	SIG	Laser	IN	TBD
Power Board	J7	Arm PD Board	PWR	TBD
Arm PD Board	24V	Arm Box Board	PWR	TBD
Power Board	J22	Science-Arm Board	IN	TBD

Gauge (AWG)	Wire Colour	Harness ID	Status	Notes
TBD	Red/Blk		DRAFT	Dual 8S packs
TBD	Red		DRAFT	
TBD	Red		DRAFT	
TBD	Black		DRAFT	
TBD	Red		DRAFT	
TBD	Black		DRAFT	
TBD	Red		DRAFT	
TBD	Red		DRAFT	
TBD	Red		DRAFT	J15 Fans silkscreen
TBD	Red		DRAFT	J16 Fans silkscreen
TBD	Yellow		DRAFT	J17 Rocker L silkscreen
TBD	Yellow		DRAFT	J18 Rocker R silkscreen
TBD	Yellow	H-DRV-01	DRAFT	
TBD	Yellow	H-DRV-01	DRAFT	
TBD	Yellow	H-DRV-01	DRAFT	
TBD	Yellow		DRAFT	
TBD	Yellow		DRAFT	
TBD	Yellow		DRAFT	
TBD	White	H-DRV-02	DRAFT	J26 Head Lights silkscreen
TBD	White	H-DRV-02	DRAFT	Middle
TBD	White	H-DRV-02	DRAFT	Edge
TBD	Grey		DRAFT	J23 RKS silkscreen
TBD	Blue		DRAFT	CAN J27 silkscreen
TBD	Red		DRAFT	J8 Network Switch silkscreen on PWB
TBD	Blue		DRAFT	
TBD	Blue		DRAFT	
TBD	Blue		DRAFT	
TBD	Grey		DRAFT	
TBD	Grey		DRAFT	
TBD	Blue		DRAFT	
TBD	Grey		DRAFT	
TBD	Grey		DRAFT	
TBD	Grey		DRAFT	
TBD	Grey		DRAFT	J7 RBPI ARM silkscreen
TBD	N/A		DRAFT	Direct GPIO header; no discrete wire
TBD	Green		DRAFT	GPS + PanTilt J16 silkscreen
TBD	Green	H-COM-01	DRAFT	
TBD	Green	H-COM-01	DRAFT	
TBD	Green	H-COM-01	DRAFT	
TBD	Orange		DRAFT	
TBD	Orange		DRAFT	

TBD	Orange		DRAFT	
TBD	Orange	H-ARM-01	DRAFT	
TBD	Orange	H-ARM-01	DRAFT	
TBD	Orange	H-ARM-01	DRAFT	
TBD	Blue		DRAFT	
TBD	Orange	H-ARM-02	DRAFT	
TBD	Orange	H-ARM-02	DRAFT	
TBD	Orange	H-ARM-02	DRAFT	
TBD	Purple	H-ARM-03	DRAFT	
TBD	Purple	H-ARM-03	DRAFT	
TBD	Purple	H-ARM-03	DRAFT	
TBD	Purple	H-ARM-03	DRAFT	
TBD	Purple	H-ARM-03	DRAFT	
TBD	Purple	H-ARM-03	DRAFT	
TBD	Purple	H-ARM-03	DRAFT	
TBD	Purple	H-ARM-03	DRAFT	
TBD	Orange		DRAFT	
TBD	Orange		DRAFT	
TBD	Grey		DRAFT	Arm/Sci J22 silkscreen

Range	Code	System
100s	PWR	Power
200s	DRV	Drive
300s	COM	Communications
400s	ARM	Arm
500s	SCI	Science
600s	—	(reserved)
700s	—	(reserved)
800s	—	(reserved)
900s	—	(reserved)

Type Flag	Meaning	
P	Power wire	24V rail, motor power, 5V to
D	Data wire	CAN, USB, Ethernet, UART

Devices / Notes	
BMS, Fuse, Kill Switches, Power Board, Buck Converters, Fans	
Splitter, Brushless Controllers, Drive Motors, Rocker, Headlights, Switches	
Jetson, Raspberry Pi, Network Switch, USB Hub, CANable, POE, Antenna,	GPS, UART Board, Pantilt
Arm PD, Arm Box, Arm Brushed, ESCs, Motors, Encoders, Limit Switches,	ToF, Laser
Science-Arm board	

Examples
o board
, PWM, encoder

Code	Silkscreen / Label	Board / Device	System
PWB	Power Board	Power Board	PWR
BMS	BMS	Battery Management Sys.	PWR
FUS	Fuse	Fuse	PWR
MKS	Kill Switch MKS	Kill Switch (MKS)	PWR
SKS	Kill Switch SKS	Kill Switch (SKS)	PWR
BK1	Buck 24→12V	Buck Converter 24→12V	PWR
BK2	Buck 24→5V	Buck Converter 24→5V	PWR
FAN	Fans	Fans	PWR
LPO	LiPo	LiPo Battery Pack	PWR
SPL	Splitter	Drive Splitter Board	DRV
BRL	Brushless L	Brushless Ctrl (Left)	DRV
BRR	Brushless R	Brushless Ctrl (Right)	DRV
BRF	Brushless F	Brushless Ctrl (Front)	DRV
MOL	Drive Motor L	Drive Motor (Left)	DRV
MOR	Drive Motor R	Drive Motor (Right)	DRV
MOF	Drive Motor F	Drive Motor (Front)	DRV
RKL	J17 Rocker L	Rocker Switch (Left)	DRV
RKR	J18 Rocker R	Rocker Switch (Right)	DRV
HD1	Headlight 1	Headlight #1	DRV
HD2	Headlight 2	Headlight #2	DRV
HD3	Headlight 3	Headlight #3	DRV
SWT	Switches / J23 RKS	Switches	DRV
JET	Jetson J6	Jetson	COM
RPI	RBPi ARM J7	Raspberry Pi	COM
HAT	Pi Hat	Raspberry Pi Hat	COM
NSW	Network Switch J8	Network Switch	COM
HUB	USB Hub	USB Hub	COM
CAN	CAN J27	CANable	COM
POE	POE	POE Injector	COM
ANT	Antenna J10	Antenna	COM
GPS	GPS1	GPS Unit 1	COM
GP2	GPS2	GPS Unit 2	COM
URT	UART	UART Board (Sonia)	COM
PAN	Pantilt	Pantilt Assembly	COM
PIT	Pitch Servo	Pitch Servo	COM
YAW	Yaw Servo	Yaw Servo	COM
TOC	ToF sensor (Pantilt)	ToF Sensor (Pantilt)	COM
APD	Arm PD Board	Arm PD Board	ARM
ABX	Arm Box Board	Arm Box Board	ARM
ABR	Arm Brushed Board	Arm Brushed Board	ARM
ESE	Elbow ESC	Elbow ESC	ARM
ESS	Shoulder ESC	Shoulder ESC	ARM

ESW	Waist ESC	Waist ESC	ARM
GRM	Gripper Motor	Gripper Motor	ARM
WPM	Wrist Pitch Motor	Wrist Pitch Motor	ARM
WRM	Wrist Roll Motor	Wrist Roll Motor	ARM
GRE	Gripper Encoder	Gripper Encoder	ARM
WPE	Wrist Pitch Encoder	Wrist Pitch Encoder	ARM
WRE	Wrist Roll Encoder	Wrist Roll Encoder	ARM
LS1	Limit Switch 1	Limit Switch 1	ARM
LS2	Limit Switch 2	Limit Switch 2	ARM
LSG	Limit Switch Gripper	Limit Switch (Gripper)	ARM
TOA	ToF sensor (Arm)	ToF Sensor (Arm)	ARM
LAS	Laser	Laser	ARM
SCA	Arm/Sci J22	Science-Arm Board	SCI